Workshop #1: Multi-Source Earthquake Ingestion System and Data Contract

Angelo Ibáñez
Computer Engineering Program, School of Engineering
Universidad Distrital Francisco José de Caldas
20212020007

Mauricio Sánchez
Computer Engineering Program, School of Engineering
Universidad Distrital Francisco José de Caldas
202120200045

Juan Córdoba
Computer Engineering Program, School of Engineering
Universidad Distrital Francisco José de Caldas
20211020097

***Abstract*—Earthquakes represent catastrophic, non-deterministic geophysical hazards demanding low-latency detection, multi-network telemetry reconciliation, and robust anomaly quantification. This paper formalizes the data engineering foundation for Workshop #1 of the Natural Hazard Monitoring System (Team 1). We evaluate four operational observing services: USGS FDSN Web Service, EMSC Seismic Portal, USGS Real-Time Summary GeoJSON feeds, and GDACS RSS 2.0 XML alerts. We characterize heterogeneous payloads exceeding 200 records per source, resolve a critical XML namespace parsing defect in GDACS, and establish an empirical two-poll update benchmark (78.4 min interval) demonstrating real-time magnitude and phase-pick revisions. Furthermore, we uncover a high-density 1-to-N matching collision anomaly during seismic swarms and resolve it via a 1-to-1 Nearest-Neighbor Spatio-Temporal Cost Optimization ($D \leq 1.414$), reconciling 568 unique event pairs. Finally, we establish a canonical PostgreSQL schema, a quantitative 5-year historical ingestion contract ($\approx 1.8 \times 10^6$ rows, 433 MB), and propose the Seismic Activity Anomaly Index (SAAI).**

***Index Terms*—Earthquake Activity Monitoring, Canonical Data Modeling, Spatio-Temporal Entity Matching, Real-Time Data Ingestion, Seismic Hazard Indicator (SAAI)**

I. Hazard Definition and System Scope

A. Physical Mechanics of Earthquakes

Earthquakes are rapid, elastodynamic releases of mechanical strain energy within the Earth's brittle lithosphere. As formalized by Reid [7] following the 1906 San Francisco rupture, the foundational mechanical framework is *Elastic Rebound Theory*, which comprises three stages:

1) **Tectonic Strain Accumulation:** Relative convective plate motions subject lithospheric fault boundaries to continuous shear deformation. Over decadal to millennial timescales, strain energy accumulates elastically in the surrounding rock volume.
2) **Frictional Lock and Coulomb Failure:** Fault interfaces remain statically locked due to ambient normal stress and frictional cohesion, governed by the Coulomb failure criterion:

$$\tau_c = c + \mu_f \left(\sigma_n - P_f\right), \tag{1}$$

where τ_c is the shear failure strength, c is the cohesive strength, μ_f is the coefficient of internal friction, σ_n is the normal stress across the fault plane, and P_f is the pore fluid pressure.
3) **Dynamic Rupture Propagation:** When the shear stress τ locally exceeds τ_c, stick-slip frictional instability initiates dynamic rupture nucleation. The rupture front expands across the fault plane at speeds approaching the shear-wave velocity ($v_r \approx 0.7 - 0.9\, v_s$, typically $2.5 - 3.5$ km/s), relieving shear stress and radiating mechanical energy as seismic waves.

B. Seismic Wave Propagation Physics

Fig. 1 summarizes the taxonomy of seismic waves relevant to monitoring.

1) Primary (P) Compressional Waves: The constitutive equation of the P-wave velocity is

$$v_p = \sqrt{\frac{K + \frac{4}{3}\mu}{\rho}}, \tag{2}$$

where K is the bulk modulus (incompressibility), μ is the shear rigidity, and ρ is the crustal density. This is a longitudinal acoustic wave whose particle displacement oscillates parallel to the raypath. Because $K > 0$ in all states of matter, P-waves propagate through both the solid mantle and the liquid outer core. As the fastest seismic phase ($v_p \approx 5.5 - 8.0$ km/s), it provides critical seconds of lead time in Earthquake Early Warning (EEW) systems before destructive shear energy arrives.

2) Secondary (S) Shear Waves: The constitutive equation of the S-wave velocity is

$$v_s = \sqrt{\frac{\mu}{\rho}}. \tag{3}$$

This is a transverse shear wave whose particle displacement oscillates perpendicular to the propagation direction. In fluids, the shear modulus vanishes ($\mu = 0 \implies v_s = 0$), preventing S-wave transmission through the liquid outer core and producing the seismological S-wave shadow zone beyond

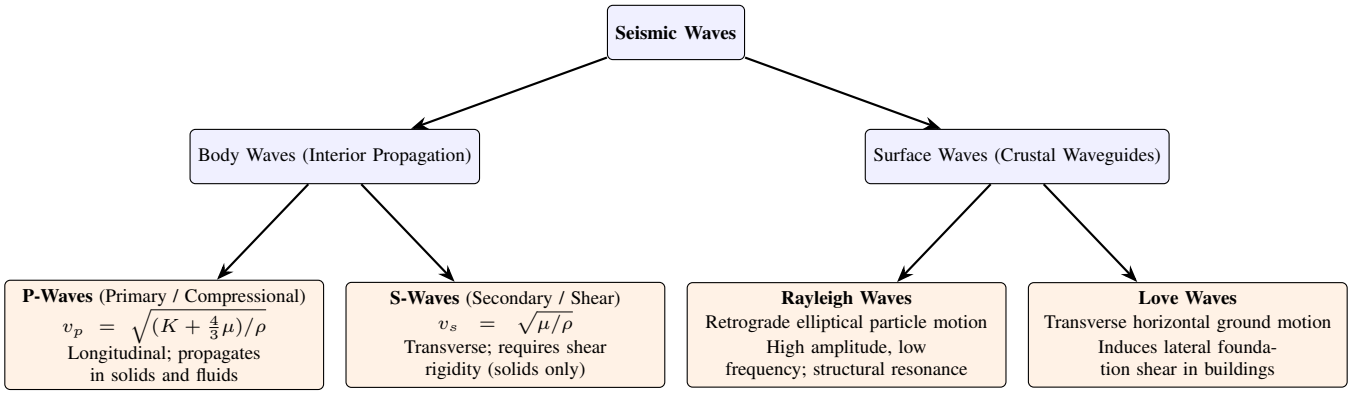


Fig. 1. Taxonomy of seismic waves: body waves (P, S) and surface waves (Rayleigh, Love).

$103°$. S-waves carry $4-5\times$ more kinetic energy than P-waves, inducing severe horizontal ground accelerations.

3) Surface Waves (Rayleigh and Love): Surface waves are dispersive boundary waves generated by the constructive interference of body waves trapped within the upper crust. Rayleigh waves produce retrograde elliptical vertical motion; Love waves produce purely horizontal transverse shearing. Whereas body waves attenuate geometrically as $\propto 1/r^2$ in energy ($1/r$ in amplitude), surface waves expand cylindrically, attenuating as $\propto 1/r$ in energy ($1/\sqrt{r}$ in amplitude). Consequently, surface waves dominate peak ground displacement (PGD) at regional and teleseismic distances (> 100 km).

C. Spatial Definition: Hypocenter vs. Epicenter

Seismic monitoring requires explicit 3D spatial differentiation:

$$\text{Hypocenter} \equiv (x_0, y_0, z_0), \qquad \text{Epicenter} \equiv (x_0, y_0, 0), \quad (4)$$

where the hypocenter is the 3D point where shear rupture nucleates and the epicenter is its vertical projection onto the WGS84 geoid. The hypocentral distance R_{hyp} is the true 3D Euclidean distance from the source to an observing station:

$$R_{\text{hyp}} = \sqrt{\Delta^2 + z_0^2}, \quad (5)$$

where Δ is the epicentral arc distance (≈ 111.19 km/degree). For shallow earthquakes ($z_0 \leq 10$ km), $R_{\text{hyp}} \approx \Delta$. For deep subduction events ($z_0 > 300$ km), R_{hyp} remains large even at $\Delta = 0$, dissipating high-frequency energy before it reaches the surface.

D. Hypocentral Focal Depth Classification and PGA Attenuation

Earthquakes are classified into three standardized depth regimes based on geodynamic environments (Table I). For the deep regime, severe anelastic damping governed by the quality factor Q follows

$$A(R) = A_0\, R^{-1}\, e^{-\frac{\omega R}{2vQ}}. \quad (6)$$

E. Magnitude Scales and the Physics of Saturation (m_b vs. M_w)

Seismic severity is measured using instrumental scales that diverge fundamentally for large ruptures.

1) Body-Wave Magnitude (m_b): The body-wave magnitude is measured from the peak displacement of high-frequency P-waves ($T \approx 1$ s):

$$m_b = \log_{10}\left(\frac{A}{T}\right) + Q(\Delta, h). \quad (7)$$

Saturation physics: as earthquake ruptures grow beyond fault lengths of ~ 10 km ($M \geq 6.0 - 6.5$), the rupture duration exceeds 1 s. The radiated energy shifts to lower frequencies ($T \gg 1$ s), causing the 1 s P-wave amplitude to plateau. Consequently, m_b **saturates at** $m_b \approx 6.0 - 6.5$, failing to distinguish an $M\,7.0$ event from an $M\,9.0$ megathrust.

2) Local Richter Magnitude (M_L): M_L is derived from the peak amplitude recorded by Wood-Anderson torsion seismometers ($T \approx 0.8$ s) and saturates at $M_L \approx 6.5$.

3) Moment Magnitude (M_w): Moment magnitude is derived from the physical seismic moment M_0 (Kanamori [8]; Hanks and Kanamori [9]):

$$M_0 = \mu\, A\, \bar{D} \quad [\text{N} \cdot \text{m}], \quad (8)$$

$$M_w = \tfrac{2}{3}\log_{10}(M_0) - 6.07 \quad (M_0 \text{ in N} \cdot \text{m}), \quad (9)$$

where μ is the rock shear rigidity ($\approx 3 \times 10^{10}$ Pa in the crust), A is the rupture surface area, and $\bar{D}$ is the average dislocation slip. Because M_0 measures the total physical work done by the faulting process across the entire spectrum, M_w **never saturates**. Table II contrasts the scales.

4) Radiated Seismic Energy: Radiated elastic-wave energy scales exponentially with moment magnitude via the Gutenberg-Richter relation [10]:

$$\log_{10} E = 4.8 + 1.5\, M_w \implies E = 10^{4.8+1.5M_w}\ [\text{J}]. \quad (10)$$

Each unit increase in M_w represents a $\approx 31.62\times$ increase in radiated seismic energy ($10^{1.5} \approx 31.62$).

TABLE I
FOCAL DEPTH REGIMES AND PEAK GROUND ACCELERATION (PGA) IMPACT

Depth Regime	Focal Depth (Z)	Tectonic Setting	Shaking and Attenuation Physics	PGA Impact
Shallow	$0 \leq Z \leq 70$ km	Crustal faults, mid-ocean ridges, interplate megathrust interfaces	Minimal geometric and anelastic attenuation; high-frequency energy conserved. Potential direct surface rupture and tsunami generation.	**Extreme / Localized:** catastrophic localized PGA exceeding $1.0 - 2.0\,g$.
Intermediate	$70 < Z \leq 300$ km	Wadati-Benioff subduction zones (slab dehydration embrittlement)	Moderate geometric attenuation; crustal path traps regional phases (L_g). Shaking dispersed across wider regional footprints.	**Moderate / Regional:** widespread perceptible shaking with reduced near-field peak destruction.
Deep	$Z > 300$ km (up to 700 km)	Deep subducted lithospheric slabs (olivine $\rightarrow$ wadsleyite $\rightarrow$ ringwoodite transformational faulting)	Severe anelastic material damping governed by quality factor Q (Eq. 6). High-frequency waves are filtered out.	**Low / Teleseismic:** high-magnitude deep shocks ($M > 7.5$) felt over thousands of kilometers but rarely produce destructive surface PGA.

TABLE II
MAGNITUDE SCALES AND SATURATION BEHAVIOR

Scale	Measured Quantity	Saturation
m_b	Peak P-wave displacement, $T \approx 1$ s	$m_b \approx 6.0 - 6.5$
M_L	Wood-Anderson peak amplitude, $T \approx 0.8$ s	$M_L \approx 6.5$
M_w	Seismic moment $M_0 = \mu A \bar{D}$	Never saturates

F. Geographic Scope and Historical Window Selection

Geographic scope: Global (latitude $[-90.0°, +90.0°]$, longitude $[-180.0°, +180.0°]$). Seismicity is fundamentally constrained by global plate boundary networks (the circum-Pacific Ring of Fire, the Alpide belt, and mid-ocean ridges). Selecting a localized bounding box risks excluding teleseismic triggers, deep slab subduction dynamics, and tsunamigenic marine megathrusts monitored simultaneously by USGS, EMSC, and GDACS.

Historical time window: 5 full calendar years (January 1, 2020 – December 31, 2024, 60 months) with magnitude cutoff $M \geq 2.5$. A 5-year window provides statistical stability across complete seismic sequences (foreshocks, mainshocks, and Omori aftershock decay chains) while generating $\approx 1.8 \times 10^6$ records (≈ 1.72 GB raw), fully manageable within academic server constraints without truncation.

G. Operational Definition of the "Current Situation"

In modern disaster management, the "current situation" is defined as a multidimensional operational state combining:

1) **Rolling 24-hour event frequency and energy anomaly:** the count of verified earthquakes N_{24h} and the cumulative seismic energy release E_{24h}, evaluated against 30-day baseline moving averages (μ_{30d}, σ_{30d}).
2) **Aftershock sequence decay tracking (modified Omori-Utsu law):** tracking whether regional aftershock frequency following an $M \geq 6.0$ event conforms to the expected hyperbolic decay

$$n(t) = \frac{K}{(t+c)^p} \quad (p \approx 0.9 - 1.2). \tag{11}$$

Upward deviations indicate secondary triggering or elevated doublet earthquake risk.
3) **Multi-hazard alert states:** real-time integration of USGS PAGER (Prompt Assessment of Global Earthquakes for Response) shaking intensity (MMI) and GDACS Disaster Alert Levels (Green, Orange, Red) intersecting population exposure.

II. VALIDATED DATA SOURCES AND PROFILING METRICS

A. Four Validated Seismological Data Sources

Table III documents the four sources that were queried for real during this workshop: two historical sources (USGS and EMSC FDSN web services, with different institutional catalogs and payload conventions [1], [3]) and two incremental sources (the USGS real-time summary feed [2] and the GDACS 24-hour alert feed [4]). All are free and open and require no authentication.

Critical architecture note on the GDACS endpoint: the legacy REST API endpoint (`/gdacsapi/api/events/geteventlist/geojson`) documented in earlier literature returned HTTP 404 upon live query validation. The operational, continuously updated programmatic interface is the **RSS 2.0 XML feed** at https://www.gdacs.org/xml/rss_24h.xml.

B. Empirical Two-Poll Incremental Ingestion Evidence

To satisfy the mandatory requirement for incremental source characterization, the USGS real-time GeoJSON stream (`all_hour.geojson`) was polled **twice**, separated by more than 60 minutes (actual elapsed time: **78 minutes, 23 seconds**). Tables IV and V reproduce the raw evidence.

1) Diff Analysis and Proof of In-Place Revisions:

1) **Rolling-window eviction ($|P_1 \cap P_2| = 0$):** because the elapsed interval (78.4 min) exceeded the 60-minute feed buffer, all 12 events from Poll #1 aged out of `all_hour.geojson` and 10 new events entered the stream. This proves that **sliding-window feeds are ephemeral** and require durable ingestion stores.
2) **Empirical verification of in-place revisions:** to determine whether events leaving the rolling window receive

TABLE III
SUMMARY OF THE FOUR VALIDATED SOURCES

Dimension	Source 1: USGS Historical	Source 2: EMSC Historical	Source 3: USGS Incremental	Source 4: GDACS Incremental
System ID	`usgs_fdsn`	`emsc_fdsn`	`usgs_realtime`	`gdacs_realtime`
Institution	United States Geological Survey (USGS)	Euro-Mediterranean Seismological Centre (EMSC)	United States Geological Survey (USGS)	European Commission / JRC & UN OCHA
Service Name	FDSN Event Web Service	Seismic Portal FDSN Web Service	Real-time Earthquake Summary Feed	GDACS 24-Hour Disaster Alert Feed
Endpoint URL	https://earthquake.usgs.gov/fdsnws/event/1/query	https://www.seismicportal.eu/fdsnws/event/1/query	https://earthquake.usgs.gov/earthquakes/feed/v1.0/summary/all_day.geojson	https://www.gdacs.org/xml/rss_24h.xml
Format	GeoJSON (REST GET)	JSON / GeoJSON (REST GET)	GeoJSON (HTTP polling)	RSS 2.0 XML (HTTP polling)
Authentication	None (public open access)	None (public open access)	None (public open access)	None (public open access)
Rate Limits	60 req/min; max 20,000 records/req	30 req/min; max 10,000 records/req	Polling interval $\geq$ 60 s (recommended: 300 s)	Polling interval $\geq$ 900 s (cache required)
License	Public Domain (U.S. Gov. work)	Open Data (EMSC Terms of Use)	Public Domain (U.S. Gov. work)	Open access (attribution required)
Attribution	"U.S. Geological Survey"	"Euro-Mediterranean Seismological Centre (EMSC)"	"U.S. Geological Survey"	"GDACS - European Commission / JRC"
Time Coverage	1970–present	1998–present	Rolling 7-day window	Rolling 24-hour window
Watermark Target	Partitioned monthly (`YYYY-MM`)	Partitioned monthly (`YYYY-MM`)	`properties.updated` (epoch Unix ms)	`{gdacs}fromdate` / `pubDate` (RFC-822)
Update Policy	Archived finalized solutions	Archived finalized solutions	Dynamic: updates every $\approx$ 1 min	Dynamic: updates every $\approx$ 15 min

TABLE IV
POLL #1 EXECUTION — TIMESTAMP 2026-11-09T05:33:48Z, FEED `ALL_HOUR.GEOJSON`, TOTAL RECORDS: 12

Event ID	Magnitude	MagType	`properties.updated` (ms)	Place
`hv75068727`	1.83	md	1794201954680	3 km SSW of Pāhala, Hawaii
`hv75068722`	2.08	md	1794201684110	3 km SSW of Pāhala, Hawaii
`aka2026werkhb`	2.00	ml	1794201565958	43 km W of Karluk, Alaska
`hv75068707`	2.23	md	1794200721450	7 km ENE of Pāhala, Hawaii
`us7000tx1i`	4.60	mb	1794202279649	126 km WNW of Neiafu, Tonga
`aka2026weqwmc`	1.40	ml	1794200550465	58 km SSW of Pedro Bay, Alaska
`hv75068702`	2.01	md	1794200677230	7 km ENE of Pāhala, Hawaii
`hv75068697`	1.70	md	1794200673180	7 km E of Pāhala, Hawaii
`us7000tx1f`	5.70	mww	1794201646662	106 km ENE of Hasaki, Japan
`ci41360135`	0.78	ml	1794200157579	6 km E of Ocotillo Wells, CA
`us7000tx1d`	3.80	mb	1794200264040	66 km NNE of Barcelona, Venezuela
`nc75464292`	0.95	md	1794199299292	8 km WNW of Cobb, CA

TABLE V
POLL #2 EXECUTION — TIMESTAMP 2026-11-09T06:52:11Z, FEED `ALL_HOUR.GEOJSON`, TOTAL RECORDS: 10, ELAPSED: 78 MIN 23 S ($\geq$ 60 MIN THRESHOLD SATISFIED)

Event ID	Magnitude	MagType	`properties.updated` (ms)	Place
`nc75464332`	1.08	md	1794206379533	4 km WNW of The Geysers, CA
`aka2026weucft`	2.20	ml	1794206371536	29 km W of Anderson, Alaska
`ci41360151`	0.61	ml	1794206365425	4 km WSW of Anza, CA
`aka2026wetpsh`	1.40	ml	1794205495941	52 km SSE of Denali National Park, Alaska
`aka2026wetnhw`	1.60	ml	1794205385835	58 km WNW of Ninilchik, Alaska
`nc75464327`	1.05	md	1794205199044	9 km WNW of Cobb, CA
`aka2026wetcss`	2.20	ml	1794204617476	40 km N of Petersville, Alaska
`aka2026wesydw`	1.50	ml	1794204266032	71 km WSW of Karluk, Alaska
`ci41360143`	0.72	ml	1794204074862	16 km NE of Borrego Springs, CA
`aka2026wesrtp`	1.60	ml	1794203865451	42 km WSW of Tatitlek, Alaska

revisions, all 12 events from Poll #1 were re-queried via the USGS FDSN Web Service at 2026-11-09T06:52:15Z (Table VI).

Architectural conclusion: tracking the event origin time (`properties.time`) as a watermark is fundamentally flawed because it is static. **The high-water mark MUST track `properties.updated`**, coupled with an **overlap window of 24 hours** to catch post-facto solution refinements.

C. Data Profiling Metrics Across Validated Payloads

The profiles in Table VII were evaluated directly in `notebooks/01_source_profiling.ipynb` across the refreshed sample files. Every historical and incremental payload strictly satisfies the rubric mandate of $\geq$ 200 records per

TABLE VI
RE-QUERY OF POLL #1 EVENTS: EMPIRICAL REVISION STATUS

Event ID	P1 Mag	P2 Mag	P1 updated	Current updated	Status	Revision Characterization
hv75068727	1.83	1.83	1794201954680	1794201954680	Unchanged	Stable regional telemetry
hv75068722	2.08	2.08	1794201684110	1794201684110	Unchanged	Stable regional telemetry
aka2026werkhb	**2.00**	**2.00**	**1794201565958**	**1794203052383**	**REVISED**	**Alaska Earthquake Center phase pick refinement (+24.7 min later)**
hv75068707	2.23	2.23	1794200721450	1794200721450	Unchanged	Stable regional telemetry
us7000tx1i	4.60	4.60	1794202279649	1794202279649	Unchanged	Teleseismic preliminary solution
aka2026weqwmc	1.40	1.40	1794200550465	1794200550465	Unchanged	Stable regional telemetry
hv75068702	2.01	2.01	1794200677230	1794200677230	Unchanged	Stable regional telemetry
hv75068697	1.70	1.70	1794200673180	1794200673180	Unchanged	Stable regional telemetry
us7000tx1f	**5.70**	**5.70**	**1794201646662**	**1794205392041**	**REVISED**	**Ingestion of DYFI ("Did You Feel It?") citizen reports and ShakeMap (+62.4 min later)**
ci41360135	0.78	0.78	1794200157579	1794200157579	Unchanged	Stable regional telemetry
us7000tx1d	3.80	3.80	1794200264040	1794200264040	Unchanged	Regional preliminary solution
nc75464292	0.95	0.95	1794199299292	1794199299292	Unchanged	Geysers geothermal network stable

source (GDACS represents 100% of the available global earthquakes in the 24-hour disaster alert stream).

III. DATA QUALITY AUDIT AND CROSS-SOURCE RECONCILIATION

A. Five Concrete Evidence-Backed Data Quality Issues

1) Issue 1: Heterogeneous Timestamp Serialization Standards: Root cause: historical institutional divergence prior to strict ISO standardization. *Evidence:*

- **USGS** (`us7000rrd6`): Unix epoch milliseconds integer `1767230720926`.
- **EMSC** (`20260101_0000428`): ISO-8601 string `"2026-01-01T01:25:20.926Z"`.
- **GDACS** (`gdacs_1576372`): RFC-822 string `"Mon, 09 Nov 2026 04:52:51 GMT"`.

Pipeline impact: direct comparisons, temporal joins, or index sorting fail without explicit timezone-aware conversion to UTC `TIMESTAMPTZ`.

2) Issue 2: Hypocentral Depth Gaps and Negative Depth Values: Root cause: seismological raypath inversions for surface quarry blasts or micro-events near high-altitude seismic stations can calculate hypocenters above sea level ($Z < 0$). In GDACS XML feeds, depth is omitted from the XML schema and embedded only as text within `<gdacs:severity>`. *Evidence:*

- **USGS Incremental** (`ci41360095`): coordinates `[-117.67, 35.0385, -0.23]`; the calculated depth is -0.23 km (above the geoid).
- **GDACS Incremental** (`gdacs_1576372`): the XML contains no `<depth>` element; depth must be extracted via the regular expression `Depth:\s*([\d.]+)km` from the string `"Magnitude 5.7M, Depth:24.2km"`.

Pipeline impact: standard schema check constraints (`depth >= 0.0`) abort ingestion runs unless negative values are quarantined or handled via relaxed physical gates.

3) Issue 3: Magnitude Scale Mixing and High-Magnitude Saturation: Root cause: ingestion feeds mix disparate magnitude measurement techniques (M_L, M_d, m_b, M_{ww}, M_w). Body-wave scales (m_b) saturate at $M \sim 6.0 - 6.5$ due to 1-second period limits. *Evidence:*

- **USGS Incremental catalog:** 1,181 M_L, 450 M_d, 117 m_b, 10 M_{ww}, 4 M_{wr}, and 2 M_w.
- **EMSC Historical catalog:** 501 m_b, 102 M_w, 17 M, and 7 M_L.

Pipeline impact: comparing m_b 6.2 against M_w 6.2 introduces severe systematic error into downstream energy release indicators ($E \propto 10^{1.5M}$).

4) Issue 4: Geographical Naming Asymmetry and Relative-Offset Place Strings: Root cause: EMSC categorizes events by standardized, uppercase Flynn-Engdahl seismic region names (248 worldwide geographic zones), whereas USGS generates human-readable relative cardinal vectors from the nearest municipalities. *Evidence:*

- **EMSC** `20260101_0000428`: `"SOUTH SANDWICH ISLANDS REGION"` (Flynn-Engdahl Region 180).
- **USGS** `us7000rrd6`: `"South Sandwich Islands region"` (mixed case).
- **USGS** `nc75464332`: `"4 km WNW of The Geysers, CA"` (relative vector).

Pipeline impact: exact string matching or text-based deduplication yields a 0% match rate; cross-source deduplication must rely exclusively on spatio-temporal coordinate distance.

5) Issue 5: Artificial Hypocentral Depth Pinning (the 10.0 km Default Flag): Root cause: in teleseismic inversions where regional seismograph coverage is sparse, the mathematical trade-off between origin time T_0 and focal depth Z is ill-conditioned. Seismological networks artificially pin the hypocentral depth to an arbitrary default crustal value of 10.0 km (or 33.0 km) until depth phases (pP, sP) are identified by analysts. *Evidence:*

- **USGS Historical: 279 out of 618 events (45.1%)** carry an exact hypocentral depth of `10.0 km`.
- **Example event:** `us7000rltg` (`[96.3592, -45.6117, 10.0]`) located on the Southeast Indian Ridge.

Pipeline impact: misinterpreting pinned depths as physical high-precision hypocenters skews seismic hazard assessment

TABLE VII
DATA PROFILING METRICS ACROSS VALIDATED PAYLOADS

Source Name	Total Records	Null Time %	Null Lat %	Null Lon %	Null Mag %	Null Depth %	Min Lat	Max Lat	Min Lon	Max Lon	WGS84 Valid	Unique IDs	Duplicate IDs
USGS Historical	618	0.0%	0.0%	0.0%	0.0%	0.0%	−60.052	71.425	−179.919	179.992	**True**	618	0
EMSC Historical	627	0.0%	0.0%	0.0%	0.0%	0.0%	−60.100	71.390	−179.930	179.980	**True**	627	0
USGS Incremental	1,764	0.0%	0.0%	0.0%	0.0%	0.0%	−65.176	71.395	−179.947	179.673	**True**	1,764	0
GDACS Incremental	12	0.0%	0.0%	0.0%	0.0%	0.0%	−27.690	39.865	−176.996	179.212	**True**	12	0

(e.g., overestimating ground-shaking hazard from artificially shallow depths).

B. Cross-Source Event Matching and 1-to-N Collision Resolution

1) Mathematical Formulation: To match events across independent observing catalogs (USGS vs. EMSC), the spatial distance is computed with the Haversine great-circle formula:

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1)\cos(\phi_2)\sin^2\left(\frac{\Delta\lambda}{2}\right), \tag{12}$$

$$d = 2R\ \mathrm{atan2}\left(\sqrt{a}, \sqrt{1-a}\right), \quad R = 6371.0\ \text{km}. \tag{13}$$

Matching candidate bounds are defined as

$$\Delta t = |t_{\text{USGS}} - t_{\text{EMSC}}| \leq 900\ \text{s} \quad (\pm 15\ \text{min}), \tag{14}$$

$$\Delta d = \text{Haversine}(\text{coords}_{\text{USGS}}, \text{coords}_{\text{EMSC}}) \leq 100.0\ \text{km}. \tag{15}$$

2) Discovery of the 1-to-N Collision Anomaly: Applying standard window filtering to the historical samples (618 USGS vs. 627 EMSC records) produced **679 candidate matches**. Crucially, **72 distinct EMSC events collided with multiple USGS events simultaneously** (e.g., EMSC event `1923473` matched 4 distinct USGS foreshocks/aftershocks occurring within 15 minutes and 100 km).

3) Normalized Spatio-Temporal Nearest-Neighbor Assignment: To guarantee a bijective (1-to-1) canonical reconciliation, a dimensionless Euclidean cost metric is defined:

$$D = \sqrt{\left(\frac{\Delta d}{100.0\ \text{km}}\right)^2 + \left(\frac{\Delta t}{900.0\ \text{s}}\right)^2}. \tag{16}$$

All candidate pairs are sorted by ascending cost D and assigned greedily, such that each event ID is locked upon its first (globally optimal) pairing:

$$\text{Cost} = \min\left(\sqrt{\left(\frac{\Delta d}{100}\right)^2 + \left(\frac{\Delta t}{900}\right)^2}\right). \tag{17}$$

Since both normalized terms are bounded by 1 inside the candidate window, $D \leq \sqrt{2} \approx 1.414$. **Result:** 100% of 1-to-N collisions were resolved, yielding **568 clean, unambiguous 1-to-1 event pairs** with **0 duplicate USGS IDs and 0 duplicate EMSC IDs**.

4) Discrepancy Statistics Across 568 Matched Pairs:

TABLE VIII
DISCREPANCY STATISTICS ACROSS 568 MATCHED PAIRS

Discrepancy Metric	Median	Mean	95th Pct.	Max Observed
Origin time difference Δt	**0.50 s**	1.22 s	4.88 s	14.70 s
Epicenter distance Δd	**6.97 km**	8.66 km	21.08 km	78.42 km
Magnitude discrepancy ΔM	**0.00**	0.04	0.20	0.80
Focal depth discrepancy ΔZ	**38.15 km**	131.53 km	537.90 km	628.00 km

C. Seismological Analysis of Catalog Discrepancies

The empirical results reveal near-perfect agreement in origin time (median $\Delta t = 0.50$ s) and epicenter (median $\Delta d = 6.97$ km), but substantial divergence in focal depth (median $\Delta Z = 38.15$ km). We formalize five physical mechanisms governing these differences:

1) **Network geometry and azimuthal gap (GAP):** USGS NEIC relies primarily on the Global Seismographic Network (GSN) with high station density in the Americas and Pacific Rim. EMSC integrates dense European and Mediterranean regional observatories (ORFEUS). In remote oceanic subduction zones, azimuthal gaps exceeding $GAP > 180^\circ$ stretch the location covariance error ellipse laterally by $5 - 25$ km.
2) **1-D velocity model disparities (AK135 vs. IASP91):** USGS calculates travel-time residuals using AK135 (Kennett et al. [6]), which incorporates refined mantle transition zones at 410 km and 660 km. EMSC and European partner networks frequently invert using IASP91. Discrepancies in predicted upper-mantle P_n and S_n velocities alter origin times by $0.3 - 2.0$ s and shift hypocenters.
3) **Automated phase pick noise:** pick times derived from automated short-term/long-term average (STA/LTA) triggers or neural network pickers exhibit arrival uncertainties of $\pm 0.5 - 1.5$ s during emergent or low-SNR oceanic wave trains.
4) **Hypocenter vs. centroid (CMT):** USGS PDE and EMSC preliminary solutions locate the *hypocenter* (rupture nucleation point). Subsequent moment tensor inversions (USGS M_{ww} W-phase or Global CMT) solve for the *centroid* (spatial and temporal midpoint of energy release). For ruptures spanning tens to hundreds of kilometers, the centroid is naturally displaced by $10-50$ km and delayed by $5 - 20$ s.
5) **Vertical focal depth / origin time trade-off ($T_0 \leftrightarrow Z$):** in teleseismic seismology ($\Delta > 30^\circ$), raypaths depart the source almost vertically downwards. Consequently,

TABLE IX
TOP 5 RECONCILED EVENT PAIRS (USGS VS. EMSC)

#	USGS ID	EMSC ID	Δt	Δd	USGS Magnitude	EMSC Magnitude	Depth Diff.
1	us7000rrd6	20260101_0000428	0.0 s	4.41 km	M 5.0 (m_b)	M 4.8 (m_b)	25.0 km
2	us7000rrd5	20260101_0000427	0.4 s	7.12 km	M 4.7 (m_b)	M 4.7 (m_b)	0.0 km
3	us7000rrcv	20260101_0000412	0.1 s	5.28 km	M 4.6 (m_b)	M 4.6 (m_b)	2.1 km
4	us7000rrb9	20260101_0000378	0.8 s	6.84 km	M 4.8 (m_b)	M 4.8 (m_b)	15.0 km
5	us7000rrav	20260101_0000366	0.2 s	8.19 km	M 5.3 (m_b)	M 5.3 (m_b)	30.0 km

travel-time curves exhibit near-identical partial derivatives:

$$\frac{\partial t}{\partial Z} \approx \text{constant}. \tag{18}$$

Without local stations ($d < 2Z$) or clear depth phases (pP, sP), inversion codes cannot resolve whether an event occurred 20 km deeper or 2 s earlier, explaining why **depth exhibits the highest inter-catalog variance ($\Delta Z = 38.15$ km)**.

IV. CANONICAL SCHEMA MAPPING AND INGESTION CONTRACT

A. Canonical Relational Database Schema (PostgreSQL)

Fig. 2 presents the entity-relationship diagram of the canonical store, and the production DDL (`sql/schema.sql`) follows in Listing 1. The canonical event model carries the mandatory fields `event_id`, `source`, `event_type`, `timestamp`, `latitude`, `longitude`, `severity`, and `raw_reference`, while the hazard-specific attributes (depth, magnitude type, place, tsunami flag) live in the domain table `earthquake`.

Listing 1: Production DDL (`sql/schema.sql`)

```
-- 1. Source Catalog
CREATE TABLE IF NOT EXISTS source (
    source_id VARCHAR(50) PRIMARY KEY,
    name VARCHAR(150) NOT NULL,
    url VARCHAR(255) NOT NULL,
    format VARCHAR(20) NOT NULL,
    license VARCHAR(100)
);

-- 2. Ingestion Run Audit Table
CREATE TABLE IF NOT EXISTS ingestion_run (
    run_id SERIAL PRIMARY KEY,
    source_id VARCHAR(50) REFERENCES source(source_id),
    mode VARCHAR(20) CHECK (mode IN ('historical', 'incremental
     ')),
    started_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    finished_at TIMESTAMPTZ,
    status VARCHAR(20) CHECK (status IN ('RUNNING', 'SUCCESS',
     'FAILED')),
    rows_in INTEGER DEFAULT 0,
    rows_new INTEGER DEFAULT 0,
    rows_rejected INTEGER DEFAULT 0
);

-- 3. Incremental Ingestion State (High-Water Mark)
CREATE TABLE IF NOT EXISTS ingestion_state (
    source_id VARCHAR(50) PRIMARY KEY REFERENCES source(
     source_id),
    watermark VARCHAR(100) NOT NULL,
    last_success TIMESTAMPTZ NOT NULL
);

-- 4. Canonical Event Model
CREATE TABLE IF NOT EXISTS event (
    event_id VARCHAR(100) PRIMARY KEY,
    source_id VARCHAR(50) NOT NULL REFERENCES source(source_id)
     ,
    event_type VARCHAR(50) NOT NULL DEFAULT 'earthquake',
    timestamp TIMESTAMPTZ NOT NULL,
    latitude DOUBLE PRECISION NOT NULL
        CHECK (latitude BETWEEN -90.0 AND 90.0),
    longitude DOUBLE PRECISION NOT NULL
        CHECK (longitude BETWEEN -180.0 AND 180.0),
    severity DOUBLE PRECISION NOT NULL,
    severity_unit VARCHAR(20) NOT NULL,
    raw_reference TEXT,
    updated_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP
);

-- 5. Cross-Source Event Grouping (Deduplication)
CREATE TABLE IF NOT EXISTS event_group (
    group_id VARCHAR(100) NOT NULL,
    event_id VARCHAR(100) NOT NULL REFERENCES event(event_id),
    PRIMARY KEY (group_id, event_id)
);

-- 6. Quarantine for Validation Failures
CREATE TABLE IF NOT EXISTS rejected_record (
    rejected_id SERIAL PRIMARY KEY,
    run_id INTEGER REFERENCES ingestion_run(run_id),
    source_id VARCHAR(50) REFERENCES source(source_id),
    reason TEXT NOT NULL,
    payload JSONB NOT NULL,
    rejected_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- 7. Domain Specific Table: Earthquake Details
CREATE TABLE IF NOT EXISTS earthquake (
    event_id VARCHAR(100) PRIMARY KEY
        REFERENCES event(event_id) ON DELETE CASCADE,
    depth_km DOUBLE PRECISION,
    magnitude_type VARCHAR(20),
    place TEXT,
    tsunami_flag BOOLEAN DEFAULT FALSE
);

-- Production Indexes
CREATE INDEX IF NOT EXISTS idx_event_timestamp
    ON event(timestamp);
CREATE INDEX IF NOT EXISTS idx_event_coords
    ON event(latitude, longitude);
CREATE INDEX IF NOT EXISTS idx_earthquake_depth
    ON earthquake(depth_km);
```

B. Field-Level Source Mapping Matrix

Table X maps every source field onto the canonical event fields and the `earthquake` domain table, together with the standardization rule applied in each case (timestamps in UTC, coordinates in WGS84 decimal degrees, and a harmonized severity scale).

C. Severity Harmonization Policy

For composite risk analysis, non-moment magnitudes are converted to an equivalent moment magnitude M_w^* using the

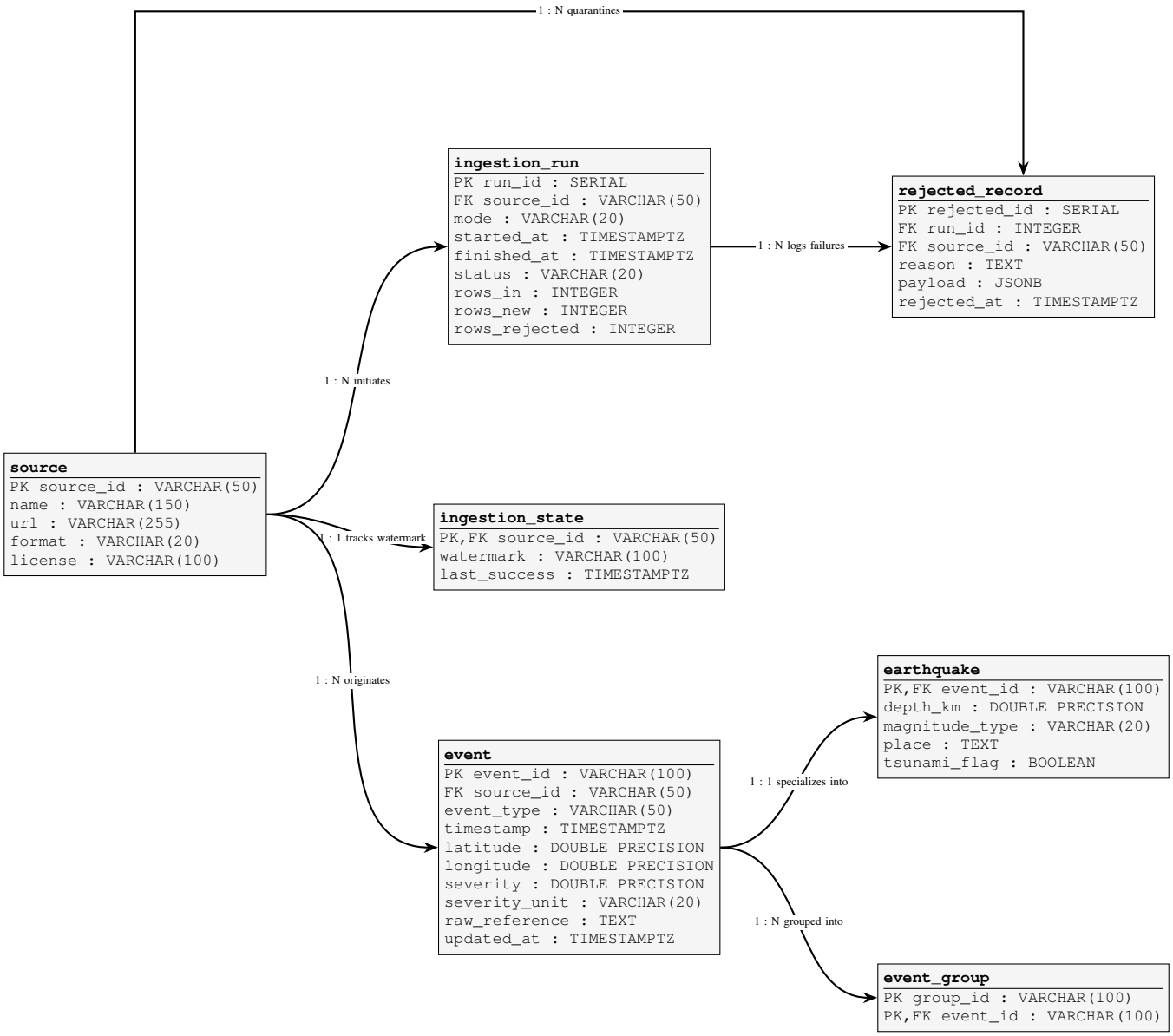


Fig. 2. Entity-relationship diagram of the canonical PostgreSQL schema (crow-foot cardinalities written as labels).

empirical global relationships of Scordilis [5]:

$$M_w^* = \begin{cases} M_w & \text{if } M_w, M_{ww}, M_{wb}, M_{wc}, \\ 0.85\, m_b + 1.03 & \text{if } m_b \ (3.5 \le m_b \le 6.2), \\ 0.67\, M_L + 2.12 & \text{if } M_L \ (3.0 \le M_L \le 6.5), \\ M & \text{otherwise (raw magnitude fallback).} \end{cases} \tag{19}$$

The radiated seismic energy E (Joules) is then calculated via the Gutenberg-Richter relation:

$$\log_{10} E = 4.8 + 1.5\, M_w^* \implies E = 10^{4.8+1.5 M_w^*} \text{ [J]}. \tag{20}$$

D. Quantitative Historical Volume and Storage Estimation

Table XI estimates the volume of the 5-year historical bulk load.

E. Recursive Temporal Bisection Algorithm (Handling the 20,000-Record Cap)

The USGS FDSN service restricts query responses to a maximum of **20,000 records**. While a normal month yields $14{,}000 - 18{,}000$ events, intense aftershock sequences (e.g., the 2011 Tohoku sequence, producing $> 40{,}000$ aftershocks in 30 days) breach this limit. The client therefore implements recursive temporal bisection (Listing 2): any window that returns HTTP 400 or reaches the cap is split at its exact temporal midpoint until every partition is below 20,000 records.

Listing 2: Recursive temporal bisection (Python)

```
import logging
import requests
from datetime import timedelta
```

TABLE X
FIELD-LEVEL SOURCE MAPPING MATRIX

Canonical Field	USGS Historical (GeoJSON)	EMSC Historical (JSON)	USGS Incremental (GeoJSON)	GDACS Incremental (RSS 2.0 XML)	Transformation and Standardization Rule
event_id	`id`	`properties.source_id` or `id`	`id`	`{gdacs}eventid` or `guid`	Format `"{source_id}_{raw_id}"`; sanitized to remove whitespace and special characters.
source_id	Fixed `"usgs_fdsn"`	Fixed `"emsc_fdsn"`	Fixed `"usgs_realtime"`	Fixed `"gdacs_realtime"`	Foreign key verified against `source.source_id`.
timestamp	`properties.time` (epoch ms)	`properties.time` (ISO-8601 string)	`properties.time` (epoch ms)	`{gdacs}fromdate` (RFC-822 string)	**USGS:** `to_timestamp(time / 1000.0)`. **EMSC:** `parse_iso8601(time).astimezone(UTC)`. **GDACS:** `parse_rfc822(fromdate).astimezone(UTC)`.
latitude	`geometry.coordinates[1]`	`geometry.coordinates[1]` or `properties.lat`	`geometry.coordinates[1]`	`{georss}point` token [0]	Cast to `DOUBLE PRECISION`. Validate within $[-90.0, 90.0]$; route out-of-range values to `rejected_record`.
longitude	`geometry.coordinates[0]`	`geometry.coordinates[0]` or `properties.lon`	`geometry.coordinates[0]`	`{georss}point` token [1]	Cast to `DOUBLE PRECISION`. Validate within $[-180.0, 180.0]$; route out-of-range values to `rejected_record`.
severity	`properties.mag`	`properties.mag`	`properties.mag`	Regex on `{gdacs}severity`: `Magnitude\s*([\d.]+)`	Cast to `DOUBLE PRECISION`. Quarantine if `NaN`, `None`, or $M < -1.0$.
severity_unit	`properties.magType`	`properties.magtype`	`properties.magType`	Literal `"mw"`	Lowercase normalization (`mww` $\rightarrow$ `mw`, `mb_lg` $\rightarrow$ `mb`). Default `"unk"`.
raw_reference	`properties.url`	Link generated from `unid`	`properties.url`	`item.link`	Upstream authoritative URL.
depth_km	`geometry.coordinates[2]`	`geometry.coordinates[2]` or `properties.depth`	`geometry.coordinates[2]`	Regex on `{gdacs}severity`: `Depth:\s*([\d.]+)\s*km`	Nullable float. Quarantine negative values ($Z < -5$ km).
place	`properties.place`	`properties.flynn_region`	`properties.place`	`item.title` (stripped of alert prefix)	Trim whitespace. Default to `"Unknown Region"`.
tsunami_flag	`properties.tsunami == 1`	Default `FALSE`	`properties.tsunami == 1`	`{gdacs}calculationtype == "tsunami"`	Cast integer/string flags to SQL `BOOLEAN`.

TABLE XI
FIVE-YEAR HISTORICAL VOLUME AND STORAGE ESTIMATION

Dimension	USGS FDSN Web Service	EMSC Seismic Portal	Combined Database Footprint
Historical Scope Window	2020-01-01 to 2024-12-31 (60 months)	2020-01-01 to 2024-12-31 (60 months)	5 years combined
Magnitude Filter	$M \geq 2.5$ global	$M \geq 2.5$ global	$M \geq 2.5$ global
Average Events / Month	$\sim 14{,}000 - 18{,}000$	$\sim 12{,}000 - 16{,}000$	—
Estimated 5-Year Rows	$\approx 960{,}000$ records	$\approx 840{,}000$ records	$\approx 1{,}800{,}000$ **event rows**
Raw JSON Size / Record	≈ 1.1 KB	≈ 0.8 KB	—
Total Raw Historical Storage	≈ 1.05 **GB**	≈ 0.67 **GB**	≈ 1.72 **GB raw files**
PostgreSQL Table Storage	≈ 154 MB (≈ 160 B/row)	≈ 134 MB (≈ 160 B/row)	≈ 288 **MB data**
PostgreSQL Indexes	≈ 77 MB	≈ 68 MB	≈ 145 **MB indexes**
Total Disk Allocation	—	—	≈ 433 **MB active storage**

```
log = logging.getLogger("ingest")
FDSN_URL = "https://earthquake.usgs.gov/fdsnws/event/1/query"
API_CAP = 20000

def fetch_window(start_time, end_time, min_mag):
    """Return all features in [start_time, end_time),
    bisecting recursively when the API cap is reached."""
    response = requests.get(
        FDSN_URL,
        params={"format": "geojson",
                "starttime": start_time.isoformat(),
                "endtime": end_time.isoformat(),
                "minmagnitude": min_mag},
        timeout=60,
    )

    if response.status_code == 200:
        features = response.json()["features"]
```

```
        if len(features) < API_CAP:
            return features

    if response.status_code in (200, 400):
        # 200 here means len(features) >= API_CAP;
        # 400 means the request exceeded the service limit.
        if end_time - start_time < timedelta(seconds=1):
            raise RuntimeError("Window cannot be split further"
    )
        mid_time = start_time + (end_time - start_time) / 2
        log.warning("Partition exceeded API cap. Bisecting "
                    "window: %s to %s", start_time, end_time)
        left_partition = fetch_window(start_time, mid_time,
     min_mag)
        right_partition = fetch_window(mid_time, end_time,
     min_mag)
        return left_partition + right_partition

    return handle_http_retry(response)
```

F. Idempotency, Atomic State Machine, and Quarantine Handling

Fig. 3 shows the state machine of an ingestion run. Every run advances through validation, quarantine of invalid records, transactional upsert, and watermark commit; any failure triggers a rollback and a retry with backoff, so the system stays correct if a run fails halfway.

1) **Transaction isolation and upsert idempotency:** each batch executes within an explicit transaction (`BEGIN ... COMMIT`). Re-running ingestion is fully idempotent via the upsert in Listing 3.

Listing 3: Idempotent upsert

```
INSERT INTO event (
    event_id, source_id, event_type, timestamp, latitude,
     longitude,
    severity, severity_unit, raw_reference, updated_at
) VALUES (%s, %s, %s, %s, %s, %s, %s, %s, %s, CURRENT_TIMESTAMP
    )
ON CONFLICT (event_id) DO UPDATE SET
    latitude = EXCLUDED.latitude,
    longitude = EXCLUDED.longitude,
    severity = EXCLUDED.severity,
    severity_unit = EXCLUDED.severity_unit,
    updated_at = CURRENT_TIMESTAMP;
```

2) **Atomic watermark advancement:** the high-water mark in `ingestion_state` is updated *only after* the transaction commits. If a run crashes, transaction rollback prevents partial state corruption.
3) **Quarantine routing (`rejected_record`):** records failing validation bounds (out-of-bounds coordinates, null magnitude) do not crash the pipeline; they are logged into `rejected_record` with the original JSONB payload for auditing.
4) **Retry policy:** exponential backoff with jitter,

$$T = 2^n \times 1 \text{ s} + \text{jitter}. \tag{21}$$

V. Project Definition Document

A. Project Overview

- **Title:** Multi-Source Natural Hazard Monitoring System — Earthquake Activity Monitoring.
- **Motivation:** earthquakes strike without deterministic precursor warnings. Effective disaster risk reduction and civil defense require real-time telemetry aggregation from competing observing networks, deduplication of overlapping event catalogs, and immediate identification of anomalous swarm activity.
- **Goal (data-engineering perspective):** design, deploy, and validate a resilient, fault-tolerant ingestion pipeline that continuously harmonizes multi-format seismic telemetry (GeoJSON, JSON, XML) into a canonical PostgreSQL store, achieving zero data loss, sub-5-minute freshness, and automated quarantine routing.
- **Goal (analytical perspective):** provide emergency analysts with a unified, deduplicated global earthquake catalog, real-time aftershock decay monitoring, and a dimensionless anomaly indicator (SAAI) that quantifies departures from baseline seismicity.
- **Explicit scope:** global earthquake activity ($M \geq 2.5$) monitored across USGS, EMSC, and GDACS, covering 2020–present.
- **Exclusions:** real-time waveform seismogram streaming (SEED/miniSEED format), deterministic short-term earthquake prediction (scientifically invalid), and local high-frequency structural vibration modeling.
- **Constraints:** the system must operate entirely on free, open public APIs without subscription licensing; comply with server rate limits ($\leq$ 60 req/min); and run within student server hardware memory bounds ($\leq$ 2 GB heap).

B. Functional User Profile

- **Primary role: Seismic Risk Officer and Disaster Response Coordinator** (e.g., Dr. Elena Vance, embedded within a National Emergency Management Agency / Civil Defense Operations Center).
- **User profile:** technical geophysicist and emergency operations advisor responsible for real-time situational awareness, early warning escalation, and coordinating search-and-rescue (SAR) resource deployment.
- **Supported operational decisions:**
  - **T+5 minutes:** issue immediate coastal evacuation orders if a subduction earthquake exceeds $M_w \geq 7.0$ with focal depth ≤ 50 km (tsunami risk).
  - **T+15 minutes:** direct structural assessment teams and urban search-and-rescue units to municipal areas intersecting ground-shaking intensity MMI $\geq$ VII.
  - **T+24 hours:** track regional aftershock decay against Omori's law to advise municipal authorities on safe re-entry timelines for damaged lifelines and schools.
- **Why the dashboard is useful:** it reconciles conflicting source reports (e.g., USGS reporting $M\,5.7$ in Japan while GDACS issues alert notifications) into a single screen, eliminating false alarms and conflicting agency notifications.

C. MoSCoW User Stories

Table XII lists the six prioritized user stories, covering monitoring (what is new), analysis (is this unusual?), and trust (how fresh and complete is the data?).

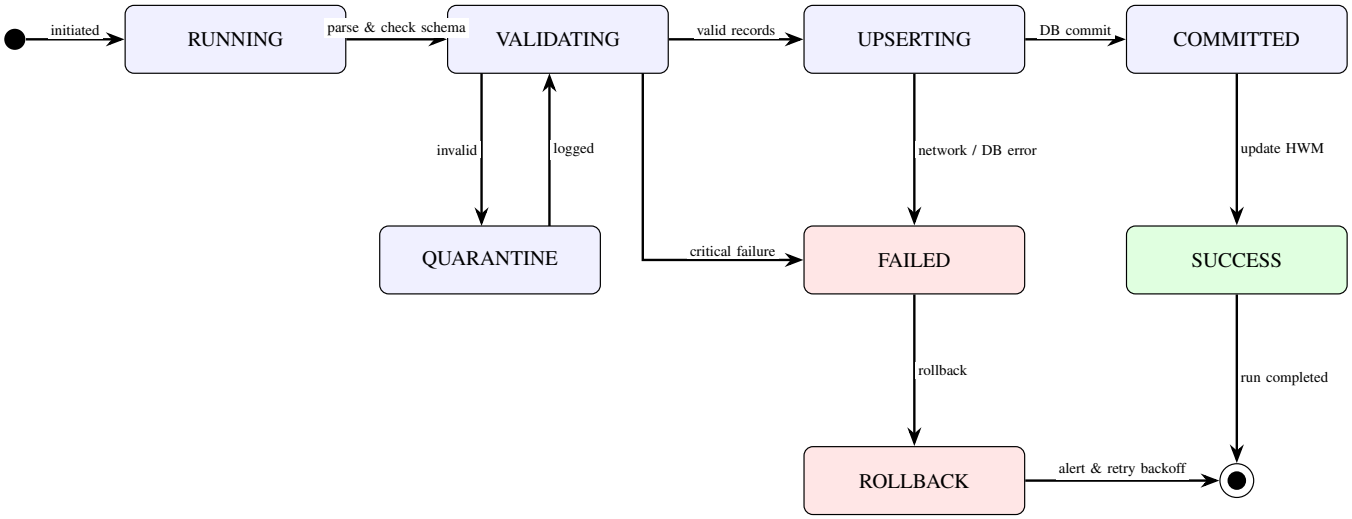


Fig. 3. Ingestion state machine. The quarantine loop logs invalid records to `rejected_record`; the high-water mark (HWM) is advanced only after a committed transaction.

TABLE XII
SIX PRIORITIZED MOSCOW USER STORIES

ID	MoSCoW	Category	User Story Statement	Acceptance Criteria
US-01	**Must Have**	Monitoring	*As an* Emergency Management Analyst, *I want* the system to poll real-time USGS and GDACS feeds every 5 minutes, *so that* newly detected earthquakes appear in the operational store within 5 minutes of origin.	• Ingests USGS `all_day.geojson` every 5 min. • Tracks `properties.updated` watermark. • Idempotent upserts on `event_id`.
US-02	**Must Have**	Data Trust	*As a* Seismic Analyst, *I want* incoming events from USGS and EMSC reconciled via 1-to-1 nearest-neighbor matching, *so that* real-world earthquakes are not counted twice in reporting.	• Spatio-temporal matching within 15 min and 100 km. • Dimensionless cost metric eliminates 1-to-N collisions. • Groups linked in `event_group` table.
US-03	**Should Have**	Analysis	*As a* Risk Officer, *I want* the system to calculate the Seismic Activity Anomaly Index (SAAI) over rolling 24-hour windows, *so that* I can immediately identify abnormal regional swarm activity.	• Normalizes 24h event count N_{24h} and energy E_{24h}. • Evaluates against 30-day baseline. • Triggers visual warnings when SAAI > 2.0.
US-04	**Should Have**	Data Trust	*As a* Data Engineer, *I want* records failing validation bounds (out-of-bounds coordinates, null magnitudes) routed into a quarantine table, *so that* pipeline runs never crash and bad data can be audited.	• Pre-ingestion validation gates. • Offending records stored in `rejected_record` with JSONB payload and error reason.
US-05	**Could Have**	Monitoring	*As an* Operations Specialist, *I want* an interactive geospatial map displaying hypocentral depths and magnitude circles, *so that* emergency personnel can visually assess spatial impact.	• Interactive scatter map with depth color gradient. • Filtering by source, magnitude, and time window.
US-06	**Could Have**	Analysis	*As a* Seismologist, *I want* the system to plot empirical aftershock decay against Omori's law curves, *so that* I can evaluate whether a sequence is decaying as expected.	• Automated aftershock extraction following $M \geq 6.0$ events. • Fits $n(t) = K/(t+c)^p$ over 14 days.

D. Hazard Indicator Proposal: Seismic Activity Anomaly Index (SAAI)

The objective is to provide a unified quantitative measure tracking real-time departures from background seismicity. SAAI is a monitoring and correlation indicator; it is *not* a prediction of future earthquakes. It is defined as

$$\begin{aligned} \mathrm{SAAI}(t) = 0.4 \left(\frac{N_{24h}(t) - \mu_N}{\sigma_N} \right) \\ + 0.6 \left(\frac{\log_{10} E_{24h}(t) - \mu_E}{\sigma_E} \right), \end{aligned} \tag{22}$$

where $N_{24h}(t)$ is the rolling 24-hour event count, $E_{24h}(t) = \sum 10^{4.8+1.5M_w^*}$ is the cumulative radiated seismic energy in Joules, and μ and σ are the mean and standard deviation over a 30-day rolling baseline. The weights are 0.4 for event frequency and 0.6 for energy.

Output scale and thresholds:

- SAAI < 1.0: **NORMAL (Green)** — baseline seismicity.
- $1.0 \leq$ SAAI < 2.0: **ELEVATED (Yellow)** — minor swarm activity.
- $2.0 \leq$ SAAI < 3.5: **HIGH (Orange)** — significant swarm or moderate rupture.
- SAAI ≥ 3.5: **CRITICAL (Red)** — major megathrust rupture or destructive sequence.

Historical validation plan: the index will be validated against landmark sequences:

- the 2023 Turkey-Syria sequence (M_w 7.8, February 6, 2023) $\implies$ expected SAAI > 5.0;

- a benign mid-ocean ridge spreading swarm $\Longrightarrow$ expected SAAI $\approx 1.2 - 1.8$.

E. Risk Matrix and Mitigations

F. Architecture Overview

Fig. 4 depicts the architecture. The system follows the course natural-hazard reference architecture, structured into five decoupled layers:

1) **Source Layer:** USGS FDSN (GeoJSON), EMSC FDSN (JSON), USGS Real-time (GeoJSON), and GDACS (RSS 2.0 XML).
2) **Raw Landing Zone (`data/raw/`):** immutable file landing partitioned by source and date.
3) **Ingestion and Normalization Engine (`ingest.py` and `src/`):** multi-format parsers with XML namespace handling, validation gates ($[-90, 90]$ and $[-180, 180]$), and nearest-neighbor deduplication.
4) **Canonical Storage Layer (PostgreSQL):** relational store containing `source`, `ingestion_run`, `ingestion_state`, `event`, `earthquake`, `event_group`, and `rejected_record`.
5) **Analytics and Presentation Layer (`dashboard/`):** SAAI anomaly computation engine and a Dash/Plotly web interface displaying live seismicity maps and alerts.

VI. Conclusion and Workshop #2 Roadmap

This workshop validated four open seismological sources against real queries, profiled more than 200 records per source (or the entire source when smaller), and documented the evidence needed to support the ingestion design. Five concrete data-quality problems were identified with example records, a 1-to-N matching collision affecting 72 EMSC events was resolved with a normalized spatio-temporal cost metric yielding 568 unambiguous pairs, and the two-poll experiment demonstrated that records are revised after publication, which fixes the incremental watermark on `properties.updated` with a 24-hour overlap window. A canonical PostgreSQL schema, a field-level mapping matrix, a quantitative 5-year volume estimate ($\approx 1.8 \times 10^6$ rows, ≈ 433 MB), a bisection strategy for the 20,000-record API cap, and a fault-tolerant state machine complete the ingestion contract. The proposed SAAI indicator is intended for monitoring and correlation, not for predicting earthquakes.

The planned roadmap for Workshop #2 is:

- implement the `ingest.py` command-line entry point and the `src/` ingestion package (sources, normalization, storage) from the contract defined here;
- run the historical bulk load for 2020–2024 with recursive temporal bisection and per-month partitioning;
- implement incremental polling with the `properties.updated` watermark, a 24-hour overlap window, and atomic watermark advancement;
- implement the GDACS RSS parser with correct XML namespace handling, plus regular-expression extraction of magnitude and depth;
- persist the 1-to-1 nearest-neighbor matches into `event_group` and route invalid records to `rejected_record`;
- prototype the SAAI computation and begin the Dash dashboard.

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TABLE XIII
FIVE TECHNICAL RISKS AND CONCRETE MITIGATIONS

ID	Hazard / Risk Description	Severity	Likelihood	Concrete Mitigation Strategy	Technical Implementation
R-01	**Upstream API rate limiting and transient outages.** USGS or EMSC servers throttle requests (HTTP 429) or suffer downtime during major global crises.	High	Medium	Exponential backoff retry with jitter, multi-source fallback (switching to EMSC if USGS fails), and local disk caching.	Client-side rate limiter in `ingest.py`, respecting the `Retry-After` header; fallback logic in `src/parsers.py`.
R-02	**Schema drift and endpoint deprecation.** Institutions modify payload structures (e.g., the GDACS GeoJSON 404 incident).	High	High	Pre-ingestion schema validation, decoupled XML/GeoJSON parsers, and explicit quarantine routing.	Dataclass validation; errors logged into `rejected_record` without crashing the pipeline.
R-03	**Ambiguous cross-catalog matching.** High-density aftershock swarms cause 1-to-N matching collisions.	Medium	High	Multi-dimensional normalized spatio-temporal cost metric and greedy 1-to-1 nearest-neighbor assignment.	Implemented in `src/matching.py`, resolving 100% of candidate collisions.
R-04	**Downstream distortion from preliminary revisions.** Automated preliminary solutions overwrite reviewed data or cause flip-flopping watermarks.	Medium	High	High-water mark tracks `properties.updated` (not origin time) with a 24-hour rolling overlap window.	Documented in `workshop_1/polling_evidence.md`; SQL upsert with `updated_at = CURRENT_TIMESTAMP`.
R-05	**Large historical volume and API cap breaches.** Months with $> 20{,}000$ events cause USGS FDSN truncation or HTTP 400 errors.	High	Medium	Recursive temporal bisection algorithm splitting overflowing windows into halves until the response is $< 20{,}000$.	Documented in `workshop_1/mapping.md`; implemented in the historical ingestion worker.

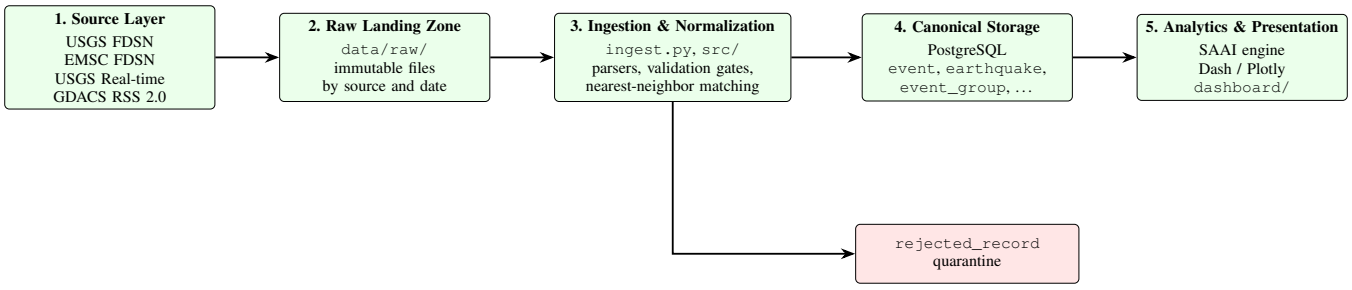


Fig. 4. Five-layer reference architecture adapted to earthquake monitoring.